

Experiment 4: Dipole Antenna Design Using HFSS

Aim

To design and simulate a dipole antenna using ANSYS HFSS and to study its S-parameter characteristics, radiation pattern, gain, and electric field distribution at an operating frequency of 5 GHz..

Software Required

- ANSYS Electronics Desktop (HFSS)

Theory:

A dipole antenna is one of the simplest and most widely used antennas in wireless communication systems. It consists of two equal-length conductive arms placed collinearly and fed at the center by a balanced source. The most common form is the half-wave dipole, whose total length is approximately $\frac{\lambda}{2}$, where λ is the wavelength of operation.

When an alternating current is applied at the feed point, currents flow in opposite directions in the two arms of the dipole. These time-varying currents produce electromagnetic radiation. The maximum current occurs at the center of the dipole and gradually decreases to zero at the ends, while the voltage is minimum at the center and maximum at the ends.

The dipole antenna radiates maximum power in the plane perpendicular to its axis and minimum power along the axis of the antenna. Hence, its radiation pattern is bidirectional and resembles a figure-of-eight in the E-plane. The radiation is linearly polarized, and the polarization depends on the orientation of the dipole.

Procedure

1. **Insert HFSS**
 - Open ANSYS HFSS
 - New Project → Insert HFSS Design
2. **Draw Dipole (Copper Cylinder)**
 - Draw → Cylinder
 - Material: **Copper**
 - Axis: Z
 - Radius: 1mm
 - Height: 12mm
 - Position: (0, 0, +0.5mm)
 - No. of segments: 0
 - Fit All (Ctrl +D)

3. **Duplicate for Second Arm**
 - Duplicate the cylinder
 - Rotate about X-axis = 180°
 - OK

4. **Dipole Formation**
 - Select both cylinders
 - Ensure double arm dipole
 - Unite
 - Change color
 - Set Transparency = 0.6

5. **Feed Creation**
 - Select YZ Plane
 - Draw → Rectangle
 - Zoom in and rotate
 - Rectangle name: Rectangle1 (Feed)

Rectangle Properties:

- Position: (0, -1, -0.5)
- Axis: X
- Y-size: 2mm
- Z-size: 1mm

(feed be placed at the center of the dipole)

6. **Assign Excitation**

- Select Rectangle1
- Excitations → Assign → Lumped Port
- Click New Line
- Assign reference conductor
- OK

(Dipole feed gap is very small and not a transmission line-hence lumped port is used)

7. **Radiation Box**

- Select XY Plane
- Create → Box
- Dimensions:
 - X = 40
 - Y = 40
 - Z = 40
- Position: $(-20, -20, -20)$
- Set Transparency = 0.8
- Assign Radiation Boundary

8. **Analysis Setup**

- Analysis → Add Solution Setup
- Frequency = 5 GHz

9. **Frequency Sweep**

- Sweep Type: Fast
- Start: 5 GHz
- Stop: 10 GHz
- Points: 451

10. **Far Field Setup**

- Results → Create Far Fields Report
- Select Radiation
- Insert Far Field Setup
- Magnitude & Phase

Angular Values:

- Phi (φ): -180° to $+180^\circ$, Step = 2°
- Theta (θ): -180° to $+180^\circ$, Step = 2°

11. **Visibility OFF**

- Turn axis / grid visibility OFF
- Used for clear plots

12. **Field Overlay**

- Results → Field Overlay

- Plot Fields → E
- Shows electric field distribution

13. **Validation & Analysis**

- HFSS → Validate Design
- Click Analyze All

14. **S-Parameter Plot**

- Results → Create Modal Solution Data Report
- Rectangular Plot
- $S(1, 1)$ in dB
- Add marker if required

15. **3D Far Field Plot**

- Results → Create Far Field Report
- 3D Polar Plot
- Quantity: Gain (Total)
- Units: dB

16. **Radiation Pattern**

- Results → Create Far Field Report
- Radiation Pattern
- Gain (Total) in dB

17. **E-Field Animation**

- Results → Animate Fields
- Select **Mag_E**
- Observe radiation behavior

Result

A dipole antenna was designed and simulated in HFSS at 5 GHz. The S11 plot shows good impedance matching, and the radiation pattern obtained is bidirectional, confirming proper dipole antenna operation.